

MECH 191 — Metallurgy

Metallography Lab Safety & PPE Quiz

Prerequisite Check — Etchants, Safety Data Sheets & Personal Protective Equipment

Fall 2026 · Leeward Community College · Instructor: Kai Santos

*Complete this quiz before beginning Module 1 hands-on lab work. Use the Etchant Safety Data Sheets, PPE Reference Guide, and EXAMET-5 Microscope & Etchant Reference pages to answer if needed — this is a comprehension check, not a memory test. **Passing score: 100% (18 of 18 correct).***

If the quiz score is sub-100% or missing, any work done within the portfolio shall be invalidated.

Student Name: _____

Student ID: _____

Semester / Year: _____

Section: _____ Date: _____

Score: _____ / 18	Result (circle): PASS FAIL	Instructor initials: _____
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Section A — Etchant & Chemical Hazard Identification

Circle the letter of the best answer.

- Q1.** Which etchant's diluent is methanol, corrected from an earlier assumption of ethanol?
- a) Marble's Reagent
 - b) Nital 2%
 - c) Kalling's No. 2
 - d) Ferric Chloride
- Q2.** Which two etchants contain hydrofluoric acid (HF)?
- a) Nital and Marble's Reagent
 - b) Keller's Reagent and Kroll's Reagent
 - c) Kalling's No. 2 and Ferric Chloride
 - d) Marble's Reagent and Kalling's No. 2
- Q3.** What specific first-aid item is called out as the skin treatment for HF exposure from Keller's or Kroll's reagent?
- a) Apply ice to the area
 - b) Apply calcium gluconate gel
 - c) Apply a baking soda paste
 - d) No further treatment is needed
- Q4.** What is the signal word on all six etchant SDS documents?
- a) Warning
 - b) Danger
 - c) Caution
 - d) Notice
- Q5.** Why is latex an unacceptable glove material for Keller's or Kroll's reagent?
- a) It dissolves on contact with water
 - b) It does not reliably block hydrofluoric acid
 - c) It is too expensive for lab use
 - d) It is uncomfortable to wear for long periods
- Q6.** Which etchant's SDS lists flammable liquid, skin corrosion, AND toxic-if-inhaled hazard classes together?
- a) Marble's Reagent
 - b) Ferric Chloride
 - c) Kalling's No. 2 Reagent
 - d) Keller's Reagent
- Q7.** Where must an HF-containing etchant be mixed and used?
- a) At any open bench
 - b) In a fume hood
 - c) Outdoors only
 - d) In the specimen storage cabinet

Q8. What does the SDS list as a hazard specific to methanol exposure that is not shared by the other diluents on this page?

- a) Skin staining
- b) Optic nerve damage / risk of blindness
- c) Permanent hearing loss
- d) Loss of sense of smell

Section B — Match Each Etchant to the Specimen Material It's Used On

Write the letter of the correct specimen material in the "Your Answer" column. Each letter is used exactly once.

Etchant	Your Answer	Specimen Material Options
1. Nital 2%	_____	A. Titanium (Ti-6Al-4V) B. Carbon steel (1018, 1045, 1075) C. Precipitation-hardening stainless (17-4PH) D. Aluminum (6061, 6063, 5052, 2024) E. Brass and copper F. Austenitic stainless steel (304, 316)
2. Marble's Reagent	_____	
3. Kalling's No. 2	_____	
4. Keller's Reagent	_____	
5. Ferric Chloride	_____	
6. Kroll's Reagent	_____	

Section C — PPE Selection

Circle the letter of the best answer.

Q9. What is the minimum eye protection required at all times in the shop, even when only observing?

- a) No eye protection needed if not touching anything
- b) ANSI Z87.1 safety glasses
- c) Reading glasses
- d) Sunglasses

Q10. What additional eye/face protection is required, beyond goggles, when using Kroll's reagent?

- a) Nothing further — goggles alone are sufficient
- b) A full face shield
- c) Sunglasses worn over goggles
- d) A welding helmet

Q11. What is the correct PPE rule when operating a lathe, mill, or drill press?

- a) Wear thick leather gloves for a better grip
- b) Wear no gloves at all
- c) Wear latex exam gloves
- d) Gloves are optional, student's choice

Q12. What glove material is correct for mixing or handling any of the six lab etchants?

- a) Latex
- b) Cotton
- c) Nitrile or neoprene
- d) No gloves are required

Q13. What must be staged at the etching station any time Keller's or Kroll's reagent is in use?

- a) A fire extinguisher only
- b) Calcium gluconate gel
- c) An eyewash bottle only, nothing else
- d) A spare pair of safety glasses

Q14. Per the welding lens shade chart, what shade range is recommended for SMAW (stick) welding above 160 amperes?

- a) 4-6
- b) 7-10
- c) 10-12
- d) 12-14

Section D — Short Answer / Applied Scenarios

Answer in complete sentences on the lines provided.

Q15. A classmate is about to etch a titanium specimen using Kroll's reagent while wearing latex exam gloves and basic safety glasses with no face shield. Identify TWO PPE errors in this scenario and explain the risk each one creates.

Q16. Explain why gloves are removed rather than worn when operating a lathe or drill press, even though gloves protect hands in most other shop contexts.

Q17. Name the specific hazard first-aid item that must be on hand any time Keller's or Kroll's reagent is in use, and explain what it treats and why.

Q18. The original portfolio PPE rule states only "safety glasses and gloves are required." Explain why this is not sufficient for every etching task in this lab, using at least one specific etchant hazard as your example.

INSTRUCTOR ANSWER KEY

Remove this page before distributing the quiz to students.

Section A

1. b — Nital 2% 2. b — Keller's Reagent and Kroll's Reagent 3. b — Calcium gluconate gel 4. b — Danger
5. b — Does not reliably block HF 6. c — Kalling's No. 2 Reagent 7. b — Fume hood 8. b — Optic nerve
damage / risk of blindness

Section B — Matching

1. Nital 2% → B (Carbon steel) 2. Marble's Reagent → F (Austenitic stainless) 3. Kalling's No. 2 → C (17-4PH)
4. Keller's Reagent → D (Aluminum) 5. Ferric Chloride → E (Brass and copper) 6. Kroll's Reagent → A
(Titanium)

Section C

9. b — ANSI Z87.1 safety glasses 10. b — Full face shield 11. b — Wear no gloves at all 12. c — Nitrile or
neoprene 13. b — Calcium gluconate gel 14. d — Shade 12-14

Section D — Expected Answer Points

15. Latex does not reliably block HF (risk of a contact burn that penetrates before it's noticed); no face shield
leaves the sides/whole face exposed to splash beyond what goggles alone stop. Award credit for identifying
both errors and the specific risk each creates, not just naming the correct PPE.

16. Gloves can be caught by a rotating spindle/chuck and pull the hand into the machine before the operator
can react — the entanglement risk outweighs the cut/abrasion protection gloves normally provide. Look for
the word "entanglement" or an equivalent description, not just "it's dangerous."

17. Calcium gluconate gel — HF is a contact poison that passes through skin with little or no immediate pain,
then damages deep tissue and pulls calcium out of the bloodstream, which a large enough contact area can
make fatal; calcium gluconate gel is the specified skin treatment for that exposure. Full credit requires
naming the item AND explaining what it responds to, not just "first aid."

18. Accept any etchant-specific example that shows "glasses and gloves" is insufficient: e.g. Keller's and Kroll's
both need specifically nitrile/neoprene gloves (not just any gloves) because of HF, and Kroll's additionally
needs a full face shield beyond goggles; Nital and Kalling's No. 2 need fume-hood ventilation and no open
flame because of methanol/ethanol flammability, which glasses and gloves don't address at all.